TOSHIBA

SPC-COM-E0061 Rev.8

SCiB™

Li-ion Rechargeable Battery for Industrial Applications

SCiB™ Industrial Pack (24V)

Model:

FP01101MCB01A

Product Specifications

Toshiba Corporation

- SCiB is a trademark of Toshiba Corporation.
- CAN is the registered trademark of ROBERT BOSCH GmbH.
- The product names mentioned in this document may be trademarks or registered trademarks of respective manufacturers.

Note

(1) The information provided in this document may not be reproduced or distributed, in whole or in part, without the prior permission of Toshiba Corporation.
(2) The information provided in this document is subject to change without notice.

Table of Contents

1. General

1.1. Scope

This document pertains to the SCiB™ Industrial Pack (24V) (hereafter called the “Battery Pack”).

1.2. Multiple Connections of SCiB™ Industrial Pack (24V)

Either one Battery Pack is to be used alone or two Battery Packs are to be connected in parallel. Note that, to prevent failure of the battery charger or connected device, it is prohibited to connect two or more Battery Packs in series, or to connect three or more Battery Packs in parallel.

Table 1-1: Allowed Number of Battery Packs Connected

		The number of Battery Packs in parallel		
		1 Pack	2 Packs	3 Packs or more
The number of Battery Packs in series	1 Pack	Allowed	Allowed	Not allowed
	2 Packs	Not allowed*1	Not allowed	Not allowed
	3 Packs or more	Not allowed	Not allowed	Not allowed

*1 If you need to connect two packs in series, the 48 V system battery pack SCiB™ Industrial Pack (48V) is available.

1.3. Glossary

Listed below are the definitions of terminologies and abbreviations used in this document.

Table 1-2: Definitions of Terminologies and Abbreviations

Item	Description
SCiB™ Industrial Pack (24V)	Battery pack consisting of 11 SCiB™ cells in series (1P11S) with a rated capacity of 22 Ah at 24 V. Model : FP01101MCB01A
SCiB™ Industrial Pack (48V)	Battery pack consisting of 11 SCiB™ cells in series (1P11S) with a rated capacity of 22 Ah. Exclusively for use in a 48-V system, two packs of SCiB™ Industrial Pack (48V) must be connected in series. Model : FP01101MCB02A
Cell	The minimum unit of batteries which makes up the Battery Pack.
1P11S	1 parallel 11 series Cell configuration of 11 series connections of one cell connected in parallel.
Battery pack	(Battery) Pack A constituent unit where multiple cells are connected and contained in a case.
BMU	Battery Management Unit A control unit that performs the protection operation of the Battery Pack, detects errors/failures, and notifies the host system of the battery capacity.
CAN	Controller Area Network Communication standard that is used to exchange data between devices interconnected.
SOC	State of Charge The charge remaining in the battery represented in percentage, where 0% indicates a completely discharged state and 100% indicates a fully charged state. SOC is sometimes expressed in the unit mAh.
No voltage contact signal	Open-Drain output signal
BOL	Beginning of Life The initial stages of the Battery Pack's life.
Self-discharge	A phenomenon in chemical batteries in which the stored charge of the battery decreases with the passage of time.
DO	Digital Output The SOC information (in 2bit, in 25-percentage increments) and the warning signal for discharge/charge/temperature are outputted by setting the output level to Open (Open-Drain) or Low.
FET	Field Effect Transistor A semiconductor switch that controls charge/discharge. (FET-ON: closed circuit, FET-OFF: open circuit)
Cell balance	The process of equalizing the voltage among the cells in the battery pack when there are cell voltage differences.

1.4. Operating Environment

Table 1-3 Operating Environment Conditions

Item	Specification
Operation environment/location	For use on AGVs, etc in a warehouse. For outdoor use, refer to the instruction manual (SPC-COM-E0063) and take appropriate measures on the side of the device where the Battery Pack is to be installed.
Operation ambient temperature	-30 to 45 deg C*1
Storage ambient temperature	-30 to 55 deg C (recommend: under 35 deg C)
Humidity	Under 85% RH (with no condensation)*2
Operation altitude	Under 2,000 m*3
Restrictions on operation environment	Avoid use or storage in the following environment. (1) A place with low/high temperature outside the environment conditions (2) A place with high humidity outside the environment conditions (3) A place with sudden temperature changes causing condensation (4) A place with a risk of splashes or submersion in water (5) A place with strong vibrations or shock (6) A place with a lot of dust (7) A place with corrosive gases (SO_2, H_2S), flammable gases, salt mist, iron mist, oil mist, etc. (8) A place near a device generating heat or with direct sunlight (9) A place near a device generating strong radio waves or magnetic field

*1. Please consult us for operation under low temperatures below 0 deg C, where the available capacity and the time needed for the open-circuit voltage to stabilize differ from those at room temperature.

*2. Condensation may damage the Battery Pack and render it inoperative.

*3. The cooling performance/withstand voltage performance of the Battery Pack degrades with increasing altitude.

(1) Cooling performance

Keep the pack cooled so that the cell temperature should not exceed 55 deg C.

(2) Withstand voltage performance

For operation at altitudes higher than 2,000 m above sea level, please consult us.

1.5. Shipping Form

The Battery Pack is to be delivered in a United Nations (UN) approved packaging for sea transportation.

In case the Battery Pack is to be transported as an individual pack via air or sea, please contact us at the contact information provided at the end of this document.

1.6. Warranty

Toshiba Corporation is to exchange without charge the product that, within one (1) year after its delivery, is confirmed to be defective in design, materials or workmanship.

Note that Toshiba Corporation shall not be liable for incidental and/or secondary damage or loss resulting from the malfunction of the product.

1.7. Restriction of Use

(1) This Battery Pack is a rechargeable industrial battery with a capacity of 2 kWh or less as defined in REGULATION (EU) 2023/1542 (EU Battery Regulation).

(2) Do not apply the Battery Pack to automobiles or aircrafts.

(3) The Battery Pack is not developed and manufactured for use on systems containing equipment[*1] directly related to human life. Do not use the Battery Pack for those systems.

(4) Special consideration[*3] is required for the operation, maintenance and management of systems[*2] containing equipment related to human safety and heavily affecting the maintenance of public functions. When the Battery Pack is to be used for those systems, contact our sales department for details.

(5) Do not use the Battery Pack in other connection configurations than those described in this document.

(6) Contact our sales department for use of the Battery Pack in other environments and conditions than those specified in this document.

*1: "Equipment directly related to human life" is as follows.
- Devices to be used in operating rooms
- Life support system
- Other equivalent medical equipment

*2: "Systems containing equipment related to human safety and heavily affecting the maintenance of public functions" is as follows.
- Operation control system and air traffic control system for collective transport system
- Main machine control system of a nuclear power plant, safety protection system of a nuclear power facility, and other lines and systems important to safety.
- Other equivalent equipment
- Electric wheelchair ...etc.

*3: The special considerations are to construct a safe system (i.e. designing a foolproof system, a fail-safe system, a redundant system, etc.) through extensive consultation with our engineers.

1.8. Disclaimer

(1) Toshiba Corporation (hereafter called "Toshiba") shall not be liable for any of the conditions described below.

- Any damage incurred by a user due to natural disasters such as earthquake, wind and flood damage, lightning and fire, salt damage, gas damage and other extraordinary natural phenomenon and actions by third party, other accident, intention or negligence of the user, misuse or use of the Battery Pack under abnormal conditions.
- Any damage incidental to the use or non-use of the Battery Pack (loss of business profit, suspension of business, change or loss of memorized contents).
- Any damage due to non-observance of the contents of this document.
- Any damage arising from the malfunction due to the operation out of specifications performed by the user.
- Any failure of the Battery Pack caused by the influence of your equipment (inverter and other equipment).

(2) The technical information described in this document is intended for informational purposes only, and does not license or warrant the intellectual property rights or other rights of Toshiba or third parties.

(3) Failure to follow the instructions or warnings in this document may compromise the safety, reliability or performance of the Battery Pack.

1.9. Reference Documents

For appropriate use of this Battery Pack, please refer to the following reference documents.

Table 1-4 Reference Documents

Item	Specification
SPC-COM-E0063	SCiB(TM) Industrial Pack (24V) Instruction Manual
SPC-COM-E0059	SCiB(TM) Industrial Pack (24V) CAN Interface Specifications

2. Product Specifications

2.1. Overview

The Battery Pack is composed of 11 SCiB™ battery cells--11 series connections of one SCiB™ cell connected in parallel (1P11S)--with BMU (Battery Management Unit) inside, which monitors the voltage, current, and temperature, and provides protection by interrupting the current.

2.2. Specifications

Table 2-1 Battery Pack Specifications

<table>
<tr><th>Item</th><th colspan="2">Specification</th><th>Notes and Applicable Standards</th></tr>
<tr><td>Model</td><td colspan="2">FP01101MCB01A</td><td>-</td></tr>
<tr><td>Designation of the Battery Pack</td><td colspan="2">TNP22/116/106[11S]M/-30+45/95</td><td>Reference
・IEC61960-3（2017）
・IEC62620（2014）</td></tr>
<tr><td>Applied cell</td><td colspan="2">SCiB™ cell</td><td>Model: NP2211F10FHB</td></tr>
<tr><td>Cell configuration</td><td colspan="2">1 in parallel 11 cells in series</td><td>-</td></tr>
<tr><td>Allowed number of packs connected</td><td colspan="2">SCiB™ Industrial Pack (24V) × 1
OR
SCiB™ Industrial Pack (24V) × 2 in parallel connection</td><td>It is prohibited to connect three or more SCiB™ Industrial Pack (24V) (Model : FP01101MCB01A) packs in parallel.
If you need to connect two packs in series, please use SCiB™ Industrial Pack (48V) (Model : FP01101MCB02A).</td></tr>
<tr><td>Nominal voltage</td><td colspan="2">25.3 V DC</td><td>-</td></tr>
<tr><td rowspan="2">Nominal capacity*1</td><td>SCiB™ Industrial Pack (24V) × 1</td><td>22 Ah, 556.6 Wh</td><td rowspan="2">Capacity measured in conditions below (one pack alone):
- Status: BOL
- Pack ambient temperature:
25±5 deg C
- Charge:
1.0 It (22 A), 0.5 It (11 A),
0.2 It (4.4 A) / cell 2.7 V-cut ; Step-down Charge
- Discharge:
0.2 It (4.4 A) / cell 1.5 V-cut</td></tr>
<tr><td>SCiB™ Industrial Pack (24V) × 2 in parallel</td><td>44 Ah, 1113.2 Wh</td></tr>
<tr><td rowspan="2">Max current (Charge/Discharge)</td><td>SCiB™ Industrial Pack (24V) × 1</td><td>125 A or less</td><td rowspan="2">- The Battery Pack can be used continuously in areas where the cell temperature does not exceed 55 deg C and the temperature of the circuit does not exceed the protection temperature.
- If it exceeds 125 A or more for 220 sec, or 210 A or more for 2 sec per pack, charging or discharging stops.
- When the Battery Pack is to be used with large current, please contact us.</td></tr>
<tr><td>SCiB™ Industrial Pack (24V) × 2 in parallel</td><td>150 A or less</td></tr>
</table>

Item	Specification		Notes and Applicable Standards
Operational voltage range	16.5 V to 29.7 V DC (Cell voltage range: 1.5 V to 2.7 V)		If the min. cell voltage remains at or under 1.9 V and uncharged for a certain period of time, the Battery Pack is shut down to reduce the risk of overdischarge. *3
External dimension	W247±2 × D188±2 × H165±2 (mm)		Dimensions per Battery Pack
Length of signal harness	250±30 (mm) × 3 harnesses		-
Mass	Approx. 8.3 kg		Mass per Battery Pack
External interface	Main circuit	Bolt nut terminal (B6)*2	Please tighten with M6 bolts and nuts of the strength class 4.8 or more. (tightening torque: 5.0 to 6.8 Nm)
	Control signal	- CAN input/output: CAN 2.0B - Contact input signal: No voltage contact signal - Contact output signal: No voltage contact signal	-
	Control signal 1	8 pins for host communication (CAN input/output, activation signal, address, etc.)	Using connectors (Please refer to section 4.2 for details) *For details of CAN output, please refer to CAN Interface Specifications (SPC-COM-E0059)
	Control signal 2	6 pins for status notification (four-level scale notification of SOC, status notification signal, 5 V DC)	Using connectors (Please refer to section 4.2 for details) No voltage contact output The maximum allowed voltage of contact output is 30 V.
	Control signal 3	8 pins for lower communication (CAN input/output, activation signal, address, etc.)	Using connectors (Please refer to section 4.2 for details) *For details of CAN output, please refer to CAN Interface Specifications (SPC-COM-E0059)
Function	Measurement	Voltage, current, temperature	-
	Action	Cell balance action	-
	Protection	- Overcharge protection - Overdischarge protection - Over current protection - Over temperature protection - Under temperature protection	-
	CAN output	- Cell voltage - Temperature inside of Battery Pack - Charge/discharge current - SOC*4 - System abnormality notification	-

<table>
<tr><th>Item</th><th colspan="2">Specification</th><th>Notes and Applicable Standards</th></tr>
<tr><td></td><td>Contact output signal</td><td>DO1: Discharge status notification
DO2: Charge status notification
DO3: Temperature status notification
DO4, DO5: SOC*4 notification on a four level scale (with 2 bit)</td><td>-</td></tr>
<tr><td>Resin case material</td><td colspan="2">Material: PC (Polycarbonate)
Color: Black</td><td>Flame retardant grade: UL94 V-0</td></tr>
<tr><td>Rated voltage and current of built-in fuse</td><td colspan="2">150 V, 200 A</td><td>Manufacturer: Mersen
Item name: A15QS200-4</td></tr>
<tr><td>Vibration</td><td colspan="2">Logarithmic sweep:
7Hz - 200Hz - 7Hz in 15 minutes:
12 sweeps;
3 mutually perpendicular axes,
No Rupture or Fire. (Amplitude: 0.8 mm (Total excursion: 1.6 mm), Peak acceleration: 8G)</td><td>UN Manual of Tests and Criteria 6th revised edition, Part III, subsection 38.3 T.3 Vibration</td></tr>
<tr><td>Shock</td><td colspan="2">Half-sine pulse:
Peak acceleration: 110 G
Pulse duration: 6 ms
3 shocks in the positive direction and 3 shocks in the negative direction in each of three mutually perpendicular axes
No Rupture or Fire</td><td>UN Manual of Tests and Criteria 6th revised edition, Part III, subsection 38.3 T.4 Shock</td></tr>
<tr><td>Dustproof and waterproof</td><td colspan="2">Equivalent to IP53</td><td>*Not IP53 guaranteed</td></tr>
<tr><td>Safety</td><td colspan="2">-</td><td>Compliant with JIS C 8715-2 (2012) (IEC 62619)
Secondary lithium cells and batteries for use in industrial applications- Part2: Tests and requirements of safety</td></tr>
<tr><td>Performance</td><td colspan="2">-</td><td>Compliant with IEC62620 (2014)
Secondary lithium cells and batteries for use in industrial applications- Part1: Tests and requirements performance</td></tr>
<tr><td>Cooling</td><td colspan="3">Self-cooling
The cell temperature should not exceed 55 deg C.</td></tr>
<tr><td>Label</td><td colspan="2">Product label
Caution label</td><td>-</td></tr>
</table>

*1: This is the initial characteristics at the time of shipping.
*2: “(B6)” is the terminal shape defined in JIS C 8702-2:2009 (IEC 61056-2:2012).
*3: It is applicable to packs of which Serial No. is started with “G123” or greater and which are produced in or after May 2019.
*4: The SOC is outputted as a reference value. The SOC estimation may be inaccurate if the battery has been operating continuously and/or at low temperature.
If you have questions, please contact us.

2.3. Specifications of Electro Magnetic Compatibility

This product has been tested under these regulations or standards as a single configuration (24V-23Ah). Depending on a customer's use case and the environment, to ensure compliance with these regulations or your local regional rules and regulations, the final configuration of your end system product may require additional compliance testing.

1) Immunity test (IEC 61000-6-2: for industrial environments)

Table 2-2 Immunity Test Specifications

Item	Specification	Notes and Applicable Standards
Electrostatic immunity	Contact discharge: ± 4kV Air discharge: ±8 kV	IEC 61000-4-2:2008 Electromagnetic compatibility (EMC)-Part 4-2: Testing and measurement techniques-Electrostatic discharge immunity test
Radiated, radio-frequency, electromagnetic field immunity	Field strength: 10 V/m (80 to 1000 MHz), 3 V/m (1.4 to 2 GHz) 1 V/m (2 to 2.7 GHz)	IEC 61000-4-3(:2006+A1:2007+A2:2010) Electromagnetic compatibility (EMC)-Part 4-3: Testing and measurement techniques-Radiated, radio-frequency, electromagnetic field immunity test
Electrical fast transient/burst immunity	For main circuit: ±2 kV/5 kHz For input-output signal data/controlling port: ±1 kV/5 kHz	IEC 61000-4-4:2012 Electromagnetic compatibility (EMC)-Part 4-4: Testing and measurement techniques-Electrical fast transient/burst immunity test
Surge immunity	For main circuit: 1.2/50 μs Open circuit test voltage, line to ground: ±0.5 kV Open circuit test voltage, line to line: ±0.5 kV	IEC 61000-4-5:2014 Electromagnetic compatibility (EMC)-Part 4-5: Testing and measurement techniques-Surge immunity test
Immunity to conducted disturbances, induced by radio-frequency fields	Voltage level: 10 V	IEC 61000-4-6:2013 Electromagnetic compatibility (EMC)-Part 4-6: Testing and measurement techniques-Immunity to conducted disturbances, induced by radio- frequency fields

2) Emission test (IEC 61000-6-4: for industrial environments)

Table 2-3 Emission Test Specifications

Item	Specification	Notes and Applicable Standards
Strength of radiated electromagnetic field	- 30 to 230 MHz Limit: 40 dB(μV/m)[QP], 10 m - 230 to 1000 MHz Limit: 47 dB(μV/m)[QP], 10 m - 1 to 2 GHz Limit: 56 dB(μV/m)[AV], 76 dB(μV/m)[PK], 3 m	IEC 61000-6-4(:2010)
Discharge emission	Conform to class-A	"Industrial, scientific and medical radio-frequency equipment – Limits and method of measurement"

2.4. Applicable Standards

This product has been tested under these regulations or standards as a single configuration (24V-23Ah). Depending on a customer's use case and the environment, to ensure compliance with these regulations or your local regional rules and regulations, the final configuration of your end system product may require additional compliance testing.

Table 2-4 Applicable Standards

Item	Applicable Standards	Notes
CE marking	Below standards and directives apply. - EMC Directive 2014/30/EC - EN 61000-6-2:2005 - EN 61000-6-4:2007+A1:2011	Applicable to packs produced in or after January 2018.
Third-party certification	･UL 62133:2015 Ed.1 ･CSA E62133 Issued: 2013/11/01 ･IEC 62133:2012	Applicable to packs with Serial No. "G122" to "G124"
	･UL 62133-2:2020 Ed.1 ･CSA C22.2#62133-2:2020 Ed.1 ･IEC 62133-2:2017	Applicable to packs with Serial No. "G125" or greater

2.5. Appearance and Dimensions

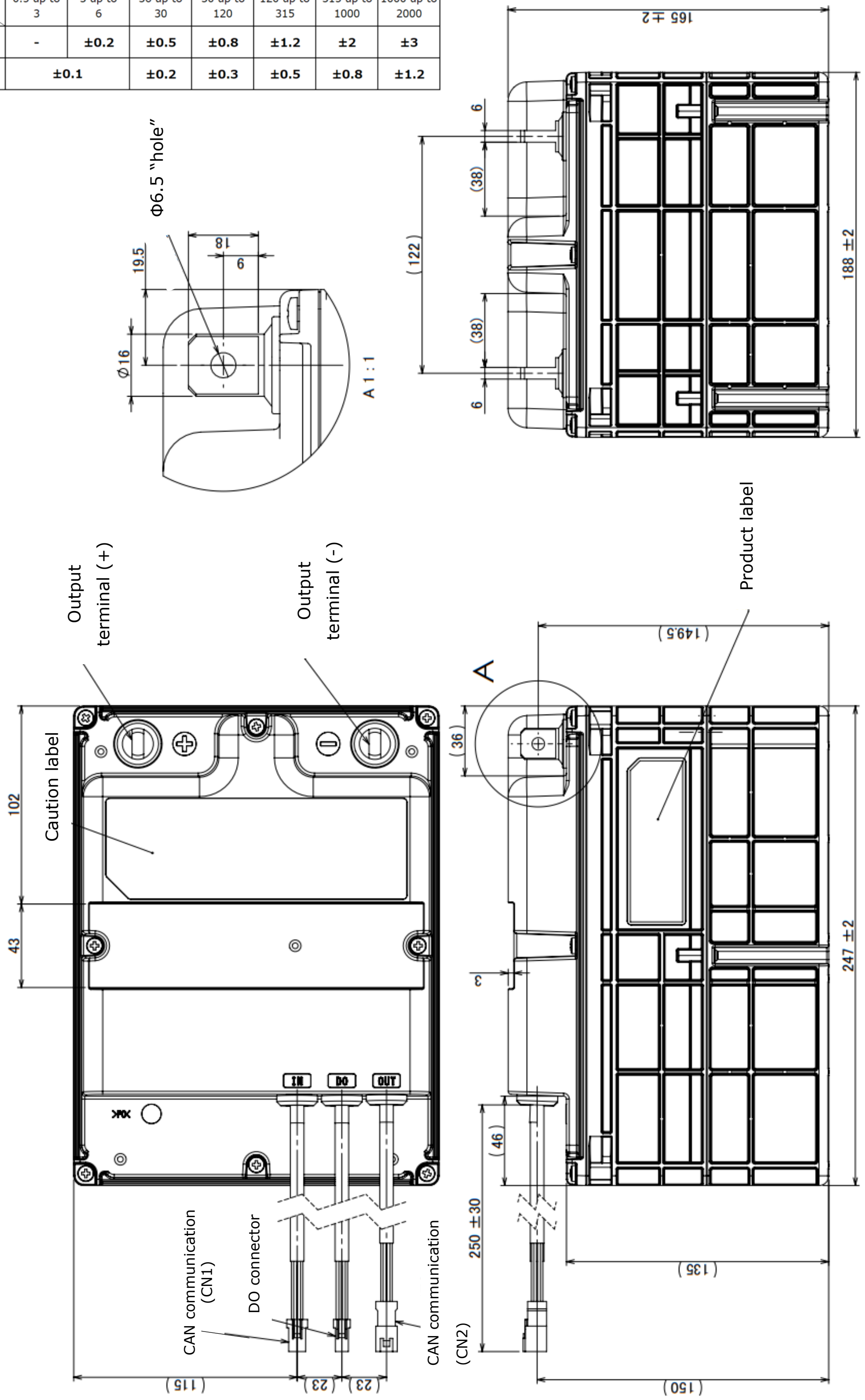
2.5.1. Appearance and Dimensions of the Battery Pack

Appearance and dimensions of the Battery Pack are shown below.

特記なき寸法公差は下記による
Unless otherwise specified,dimensional tolerances are as follows.

寸法の区分 Tolerance Class / 材質 Material	0.5 以上 3 以下 0.5 up to 3	3 を超え 6 以下 3 up to 6	6 を超え 30 以下 36 up to 30	30 を超え 120 以下 30 up to 120	120 を超え 315 以下 120 up to 315	315 を超え 1000 以下 315 up to 1000	1000 を超え 2000 以下 1000 up to 2000
非金属 Nonmetal	-	±0.2	±0.5	±0.8	±1.2	±2	±3
金属 Metal	±0.1		±0.2	±0.3	±0.5	±0.8	±1.2

(unit: mm)



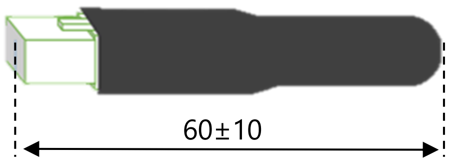
Dimensions in ( ) are reference values.

Figure 2-1 Appearance and Dimensions of the Battery Pack (Reference Image)

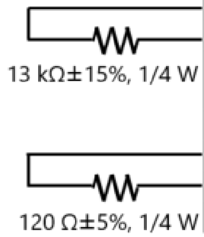
2.5.2. Appearance and Dimensions of the Termination Adapter

Appearance and dimensions of the Termination Adapter (model type: FMW-GAA0059P) packaged with the Battery Pack are shown below.

(unit: mm)



Terminator installation position (Reference Image)
If you prepare the termination adapter, refer to the image below.
Connector type：JST connector 08R-JWPF-VSLE-D



Terminal No.	Signal Name
1	MOD1_in
2	MOD2_in
3	ADR1_in
4	EN_in_Ref
5	CAN_Ref
6	CAN_L
7	CAN_H
8	EN_in_Sig

Figure 2-2 Appearance and Dimensions of Termination Adapter (Reference Image)

2.5.3. Appearance and Dimensions of the Termination Adapter 2

Termination Adapter 2 (model type: FMW-GAA0064P) is a signal harness with a built-in terminator, and is necessary when connecting two SCiB™ Industrial Pack (24V) Battery Packs in parallel without CAN communication.

(Refer to section 3.2 for details on how to connect the adapter.)

(unit: mm)

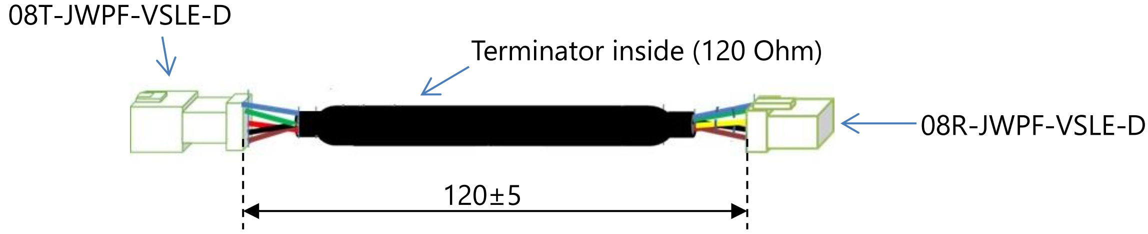


Figure 2-3 Appearance and Dimensions of Termination Adapter 2 (Reference Image)

2.6. Name of Each Part

The names of Battery Pack parts are shown below.

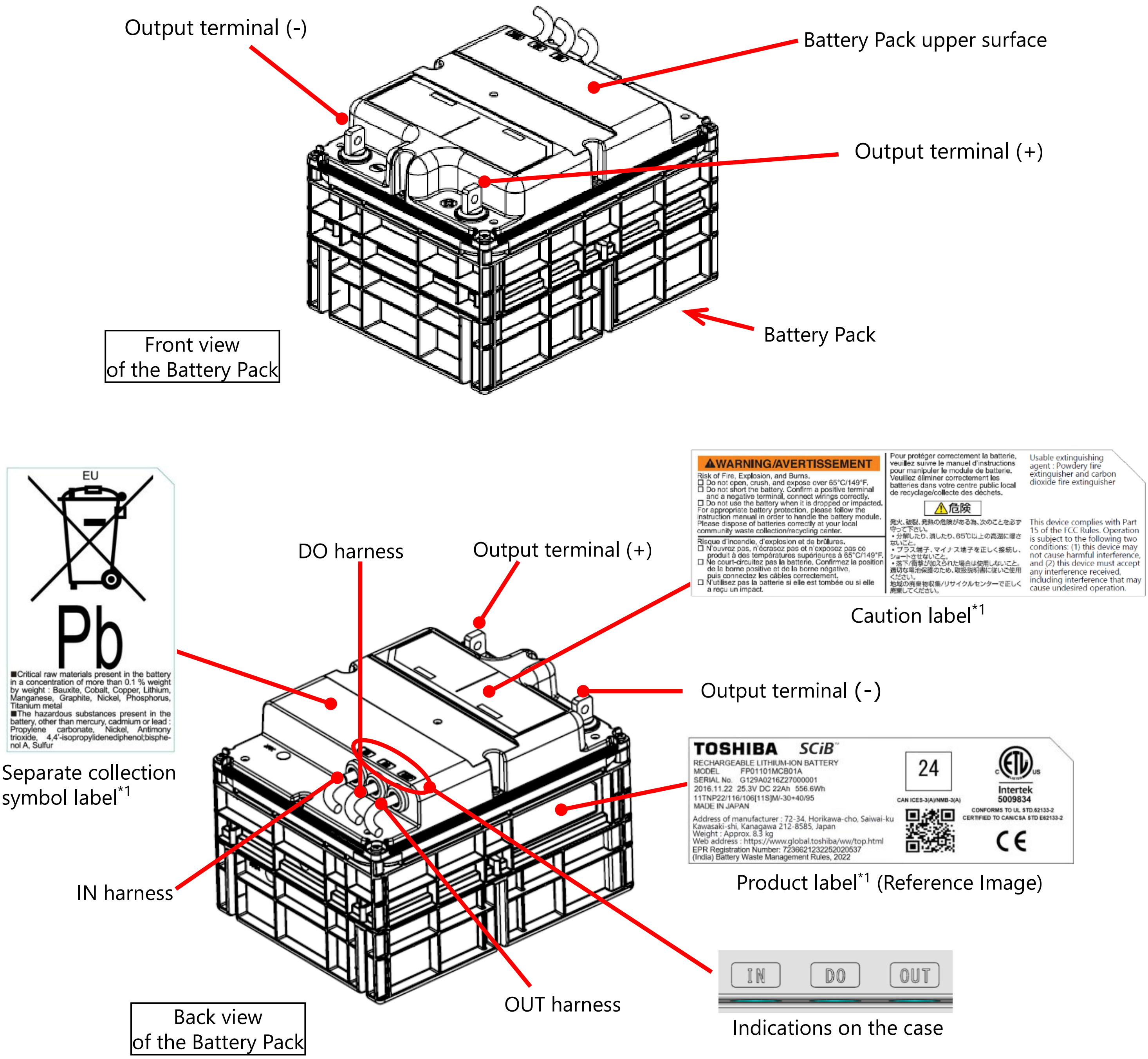


Figure 2-4 The Names of Battery Pack Parts

*1 The caution label, product label, and separate collection symbol label above are those pasted on the packs produced in or after September 2025.

3. Connection Configuration

The connection configuration of the Battery Pack, the load (your device), and the charger depends on the number of the connected battery packs (alone or 2 packs in parallel) and the communication system in use. Details for each configuration are provided in the following sections as shown on the tables below.

1) For Use of 1 Battery Pack Alone

No.	Communication System		Section	Page	Connection Configuration Figure	How to connect
	CAN Communication	DO Output				
1	Not Used	Not Used	3.1.1	14	Figure 3-1	Table 3-1
2	Used	Not Used	3.1.2	15	Figure 3-2	Table 3-2
3	Not Used	Used	3.1.3	16	Figure 3-3	Table 3-3
4	Used	Used	3.1.4	17	Figure 3-4	Table 3-4

2) For Use of 2 Battery Packs in Parallel

No.	Communication System		Section	Page	Connection Configuration Figure	How to connect
	CAN Communication	DO Output				
1	Not Used	Not Used	3.2.1	18	Figure 3-5	Table 3-5
2	Used	Not Used	3.2.2	19	Figure 3-6	Table 3-6
3	Not Used	Used	3.2.3	20	Figure 3-7	Table 3-7
4	Used	Used	3.2.4	21	Figure 3-8	Table 3-8

* For the connection procedure of the Battery Pack and the load, please refer to the instruction manual (SPC-COM-E0063).

3.1. For Use of 1 Battery Pack Alone

Sections 3.1.1 through 3.1.4 describe the connection configurations for the use of 1 Battery Pack alone.

3.1.1. With No CAN Communication or DO Output

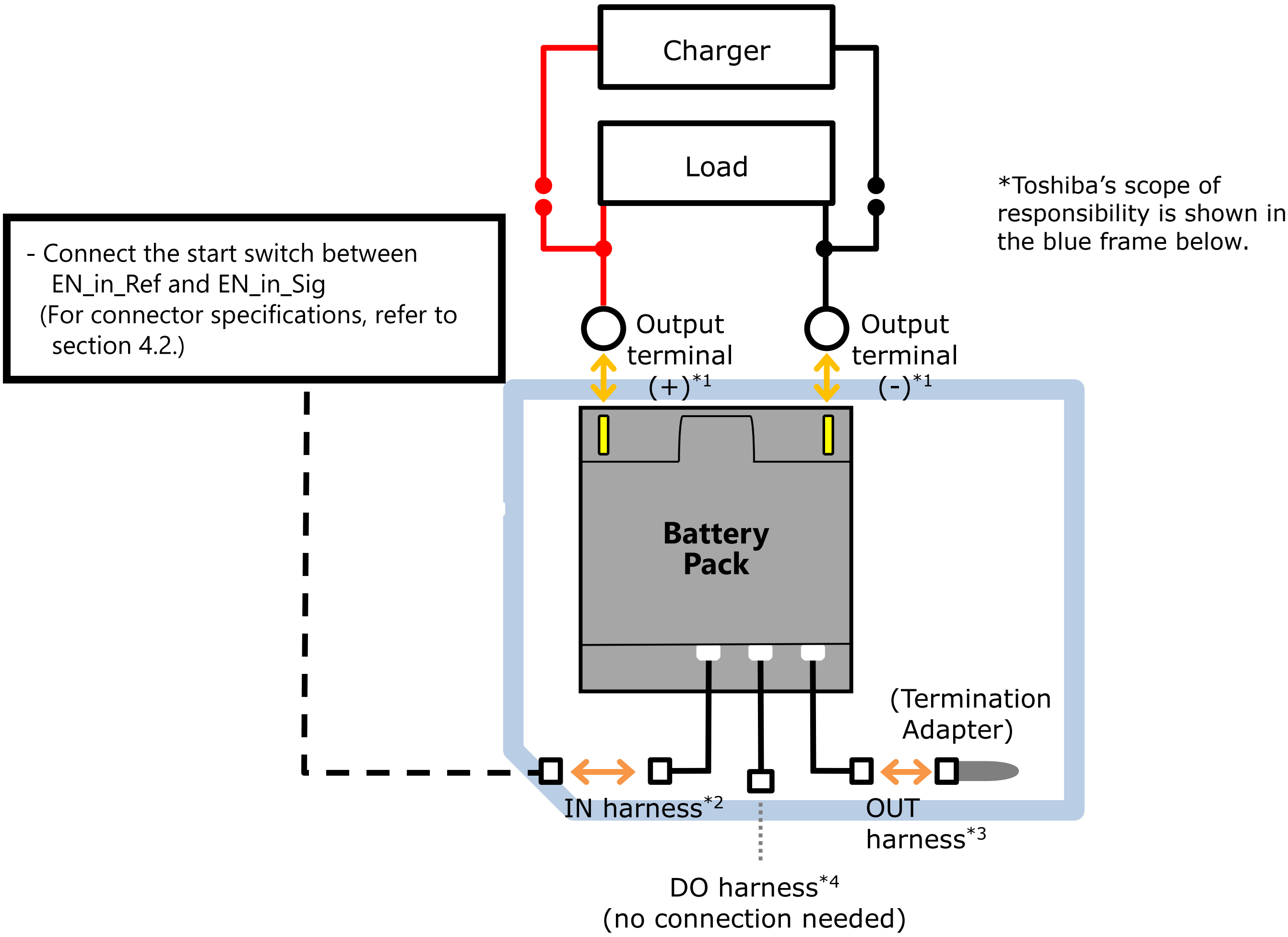


Figure 3-1 Connection Configuration with No CAN Communication or DO Output

Table 3-1 How to Connect to the Terminal/Connector

	Item	How to connect	Note
*1	Output terminal (+) Output terminal (-)	Tighten with M6 bolts and nuts the cable with M6 round terminal.	Please prepare cables with round terminal and M6 bolts and nuts.
*2	IN harness	Connect harness with connector.	-Necessary to start the Battery Pack even without connection with CAN -Please prepare the harness. Refer to section 4.2 for the connector specifications.
*3	OUT harness	Connect termination adapter.	-
*4	DO harness	No connection needed	-

3.1.2. With Only CAN Communication

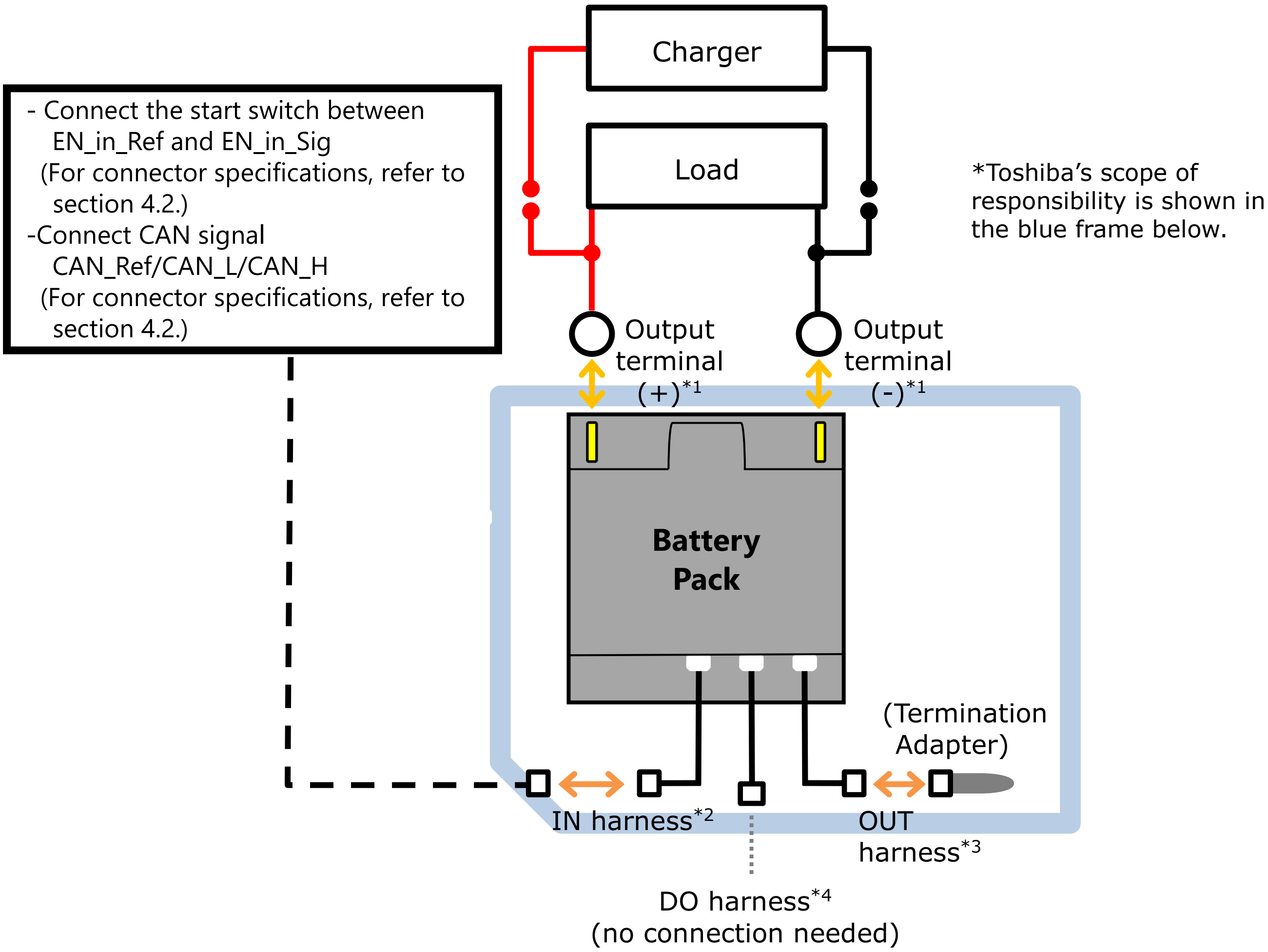


Figure 3-2 Connection Configuration with Only CAN Communication

Table 3-2 How to Connect to the Terminal/Connector

	Item	How to connect	Note
*1	Output terminal (+) Output terminal (-)	Tighten with M6 bolts and nuts the cable with M6 round terminal.	Please prepare cables with round terminal and M6 bolts and nuts.
*2	IN harness	Connect harness with connector.	Please prepare the harness. Refer to section 4.2 for the connector specifications.
*3	OUT harness	Connect termination adapter.	-
*4	DO harness	No connection needed	-

3.1.3. With Only DO Output

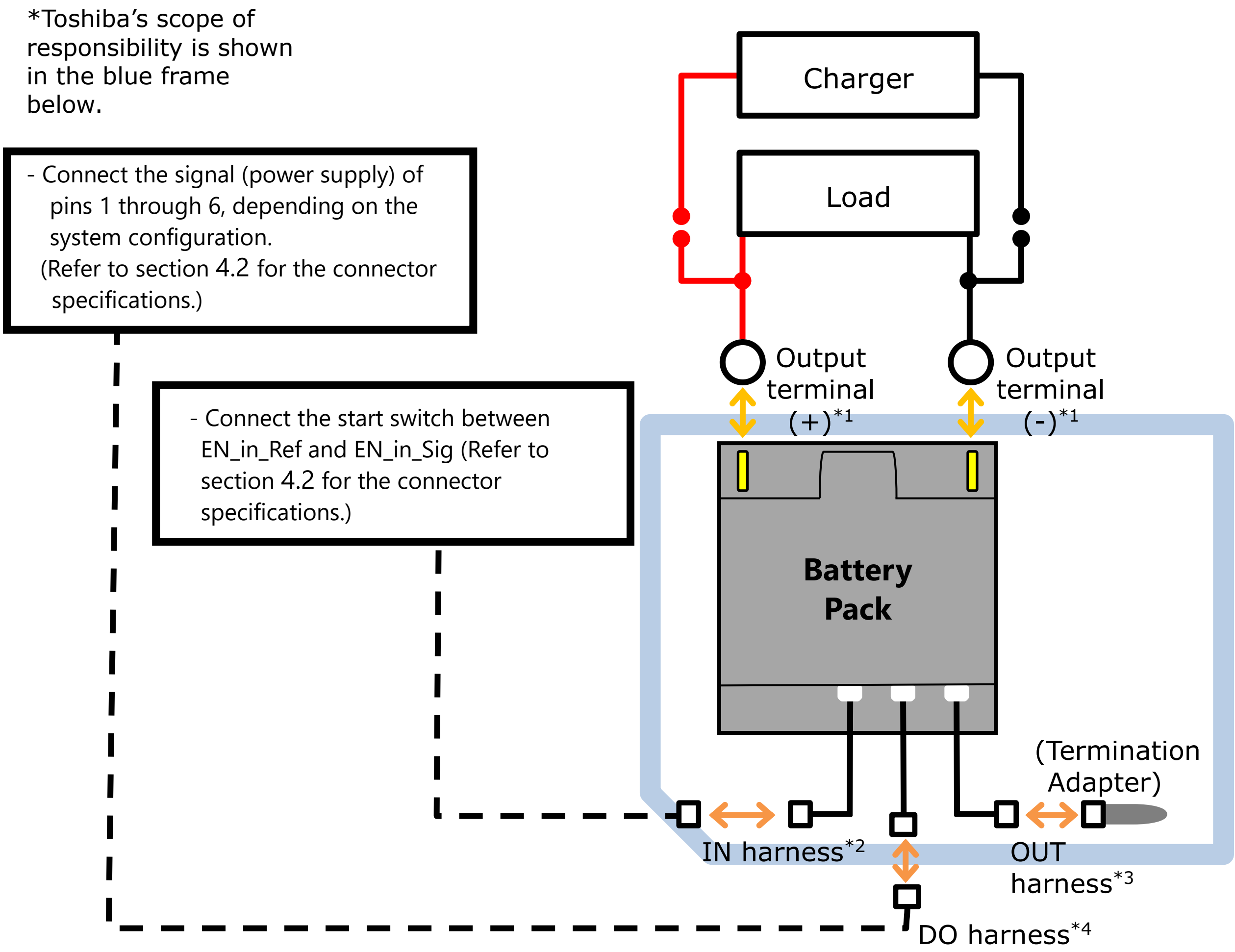


Figure 3-3 Connection Configuration with Only DO Output

Table 3-3 How to Connect to the Terminal/Connector

	Item	How to connect	Note
*1	Output terminal (+) Output terminal (-)	Tighten with M6 bolts and nuts the cable with M6 round terminal.	Please prepare cables with round terminal and M6 bolts and nuts.
*2	IN harness	Connect harness with connector.	-Necessary to start the Battery Pack even without connection with CAN -Please prepare the harness. Refer to section 4.2 for the connector specifications.
*3	OUT harness	Connect termination adapter.	-
*4	DO harness	Connect harness with connector.	Please prepare the harness. Refer to section 4.2 for the connector specifications.

3.1.4. With Both CAN Communication and DO Output

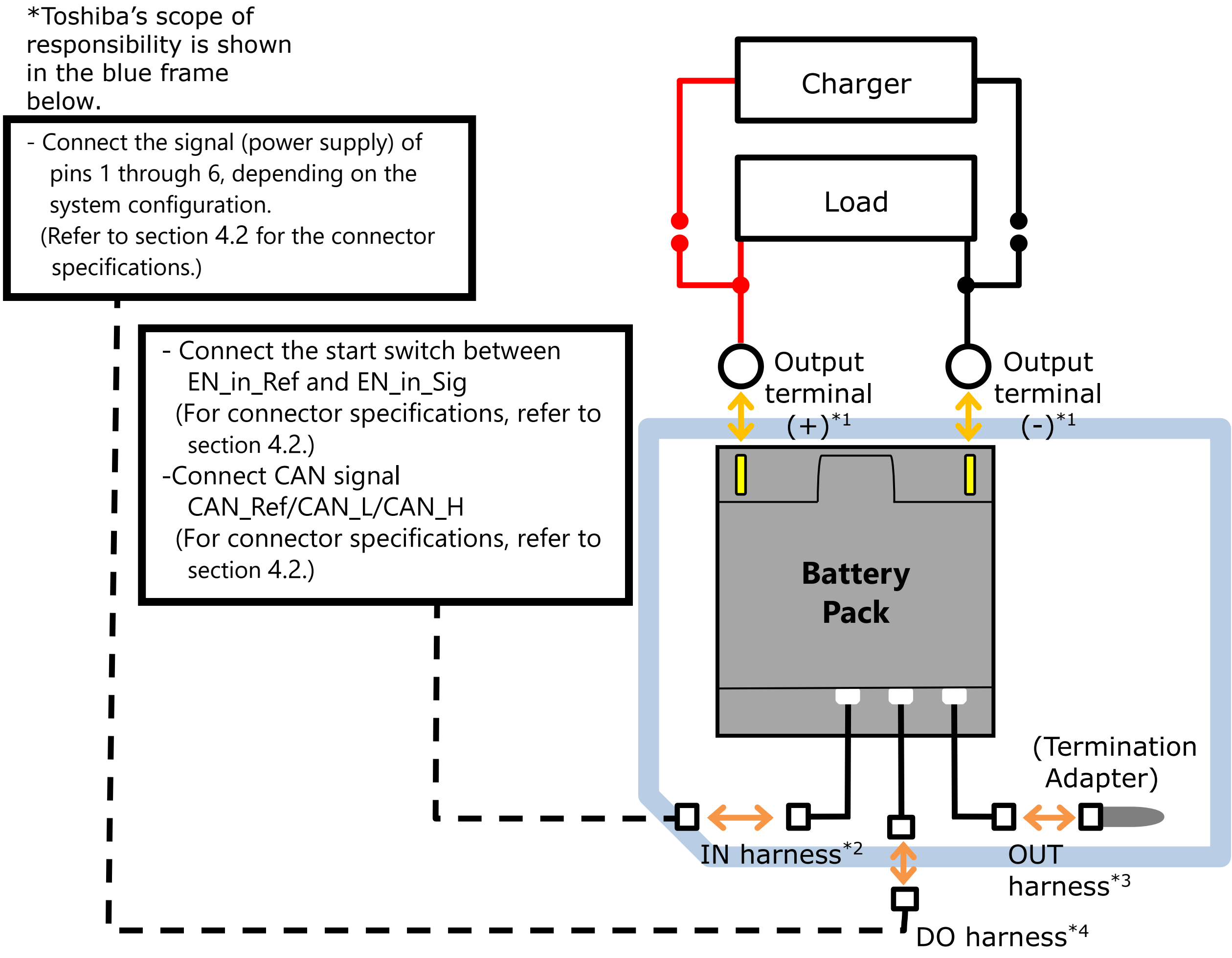


Figure 3-4 Connection Configuration with Both CAN Communication and DO Output

Table 3-4 How to connect to the Terminal/Connector

	Item	How to connect	Note
*1	Output terminal (+) Output terminal (-)	Tighten with M6 bolts and nuts the cable with M6 round terminal.	Please prepare cables with round terminal and M6 bolts and nuts.
*2	IN harness	Connect harness with connector.	Please prepare the harness. Refer to section 4.2 for the connector specifications.
*3	OUT harness	Connect termination adapter.	-
*4	DO harness	Connect harness with connector.	Please prepare the harness. Refer to section 4.2 for the connector specifications.

3.2. For Use of 2 Battery Packs in Parallel

Sections 3.2.1 through 3.2.4 describe the connection configurations for the use of 2 Battery Packs connected in parallel.

3.2.1. With No CAN Communication or DO Output

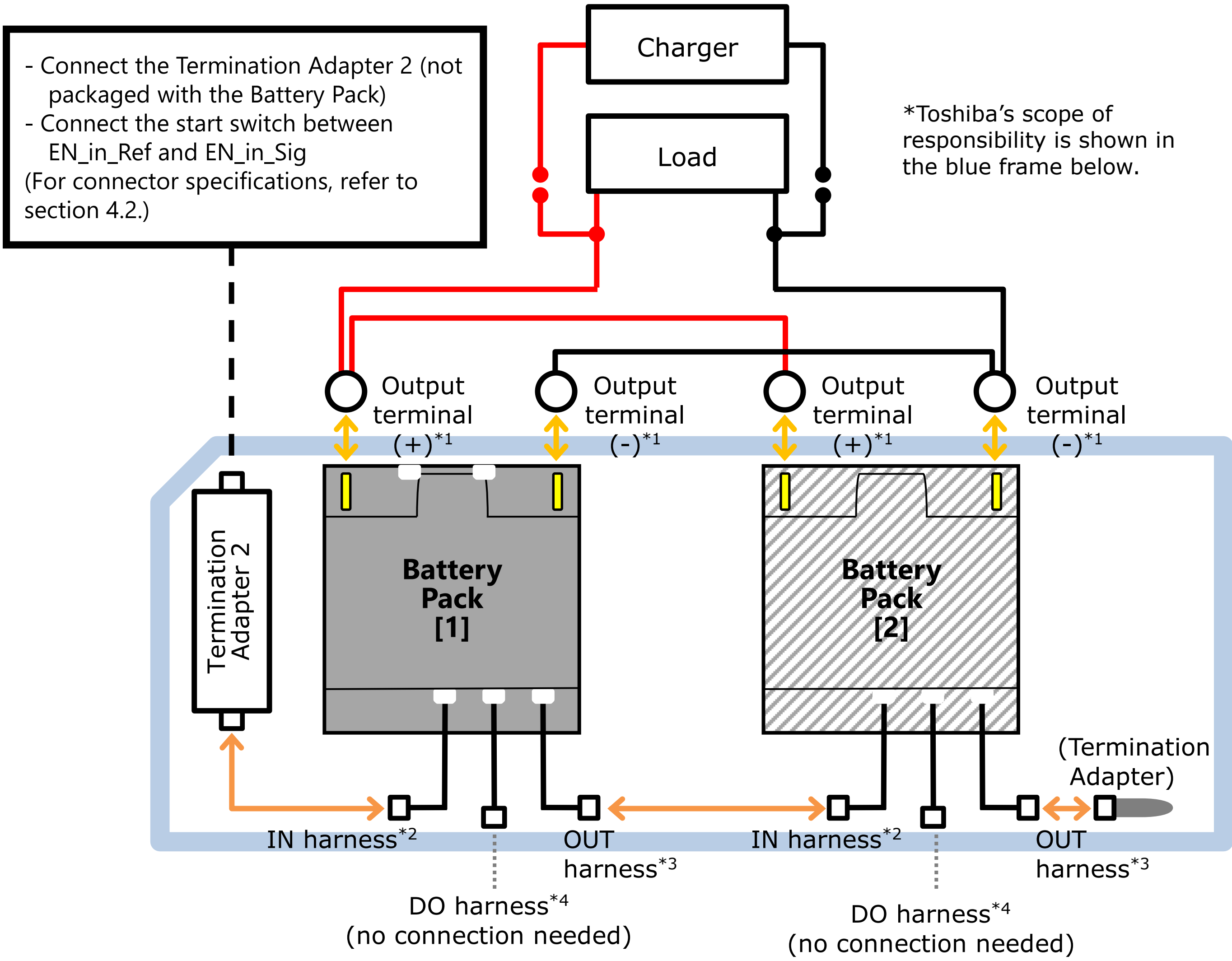


Figure 3-5 Connection Configuration with No CAN Communication or DO Output

Table 3-5 How to Connect to the Terminal/Connector

	Item	How to connect	Note
*1	Output terminal (+) Output terminal (-)	Tighten with M6 bolts and nuts the cable with M6 round terminal.	Please prepare cables with round terminal and M6 bolts and nuts.
*2	IN harness	Connect harness with connector via termination adapter 2.	Battery Pack [1] -Connect the battery pack start switch (or an equivalent circuit). -Please prepare the harness to be connected to termination adapter 2. (Refer to section 4.2 for the connector specifications.) Battery Pack [2] -Connect to OUT harness of Battery Pack [1].
*3	OUT harness	Connect termination adapter.	-
*4	DO harness	No connection needed	-

Note: Refer to section 5.4. Protection Function For 2 Packs in Parallel.

3.2.2. With Only CAN Communication

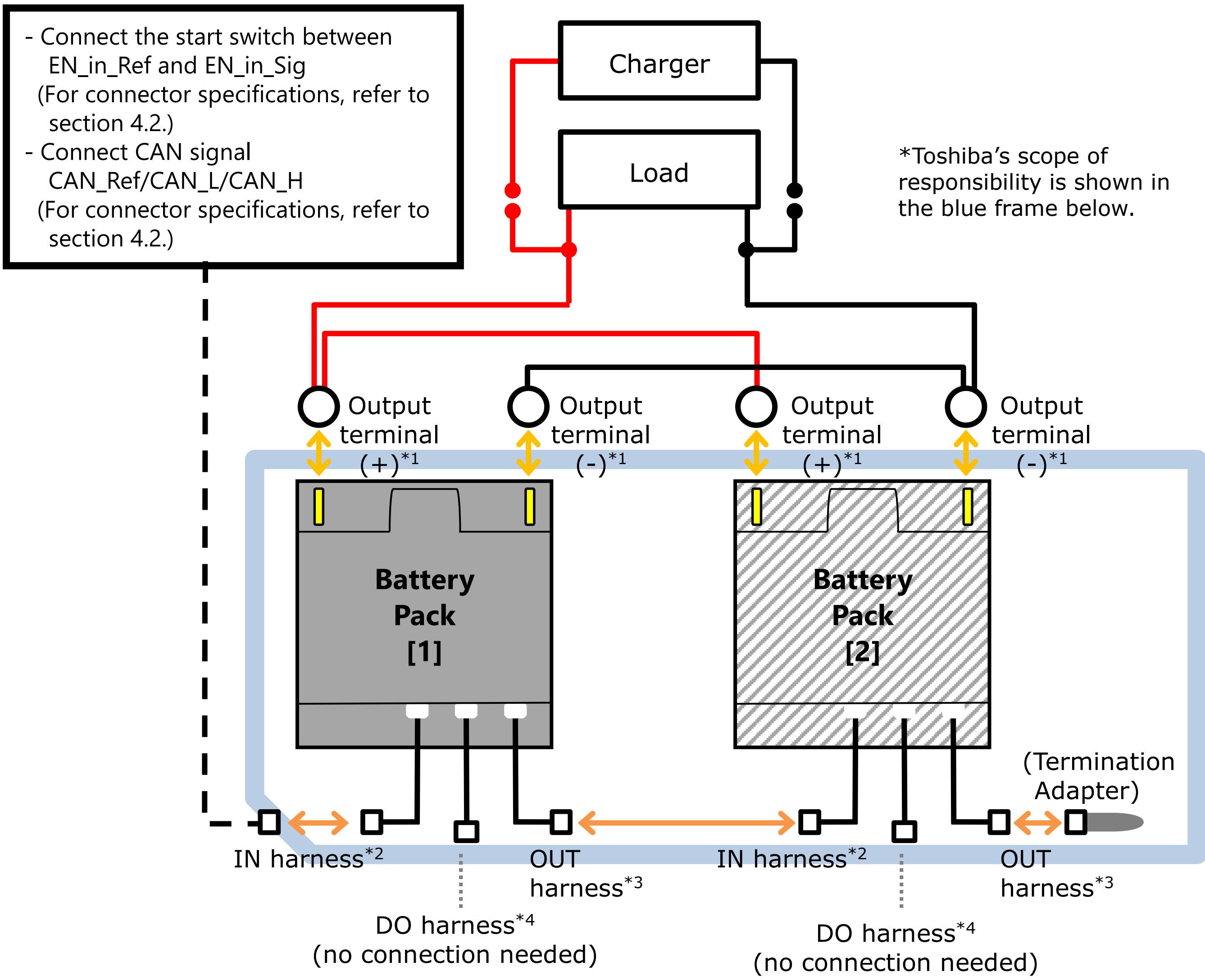


Figure 3-6 Connection Configuration with Only CAN Communication

Table 3-6 How to Connect to the Terminal/Connector

	Item	How to connect	Note
*1	Output terminal (+) Output terminal (-)	Tighten with M6 bolts and nuts the cable with M6 round terminal.	Please prepare cables with round terminal and M6 bolts and nuts.
*2	IN harness	Connect harness with connector.	Battery Pack [1] -Please prepare the harness to be connected to termination adapter 2. (Refer to section 4.2 for the connector specifications.) Battery Pack [2] -Connect to OUT harness of Battery Pack [1].
*3	OUT harness	Connect termination adapter.	-
*4	DO harness	No connection needed	-

Note: Refer to section 5.4. Protection Function For 2 Packs in Parallel.

3.2.3. With Only DO Output

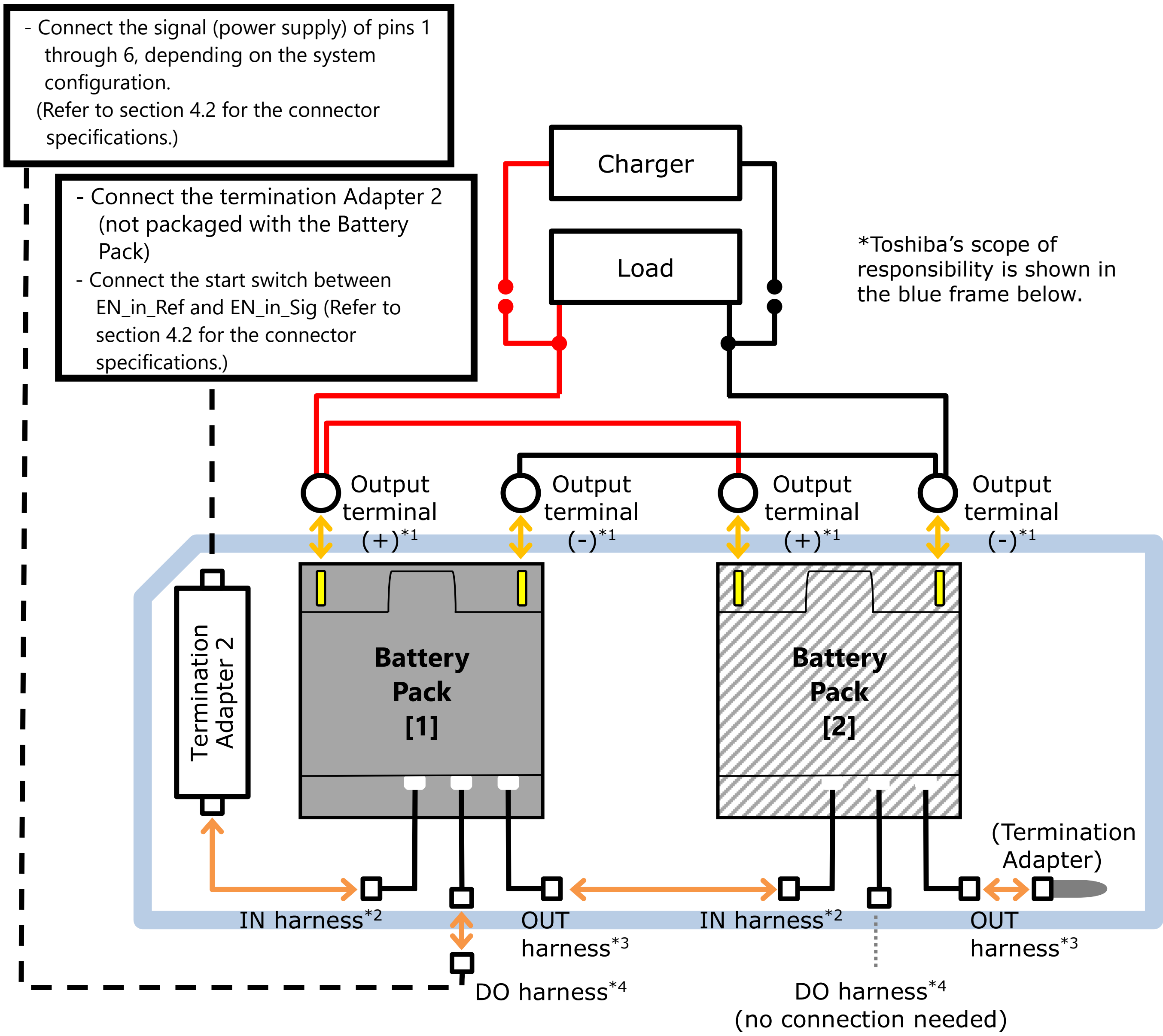


Figure 3-7 Connection Configuration with Only DO Output

Table 3-7 How to Connect to the Terminal/Connector

	Item	How to connect	Note
*1	Output terminal (+) Output terminal (-)	Tighten with M6 bolts and nuts the cable with M6 round terminal.	Please prepare cables with round terminal and M6 bolts and nuts.
*2	IN harness	Connect harness with connector via termination adapter 2.	Battery Pack [1] -Connect the battery pack start switch (or an equivalent circuit). -Please prepare the harness to be connected to termination adapter 2. (Refer to section 4.2 for the connector specifications.) Battery Pack [2] -Connect to OUT harness of Battery Pack [1].
*3	OUT harness	Connect termination adapter.	-
*4	DO harness	Connect harness with connector. *Battery Pack [1] only	Please prepare the harness. Refer to section 4.2 for the connector specifications.

Note: Refer to section 5.4. Protection Function For 2 Packs in Parallel.

3.2.4. With Both CAN Communication and DO Output

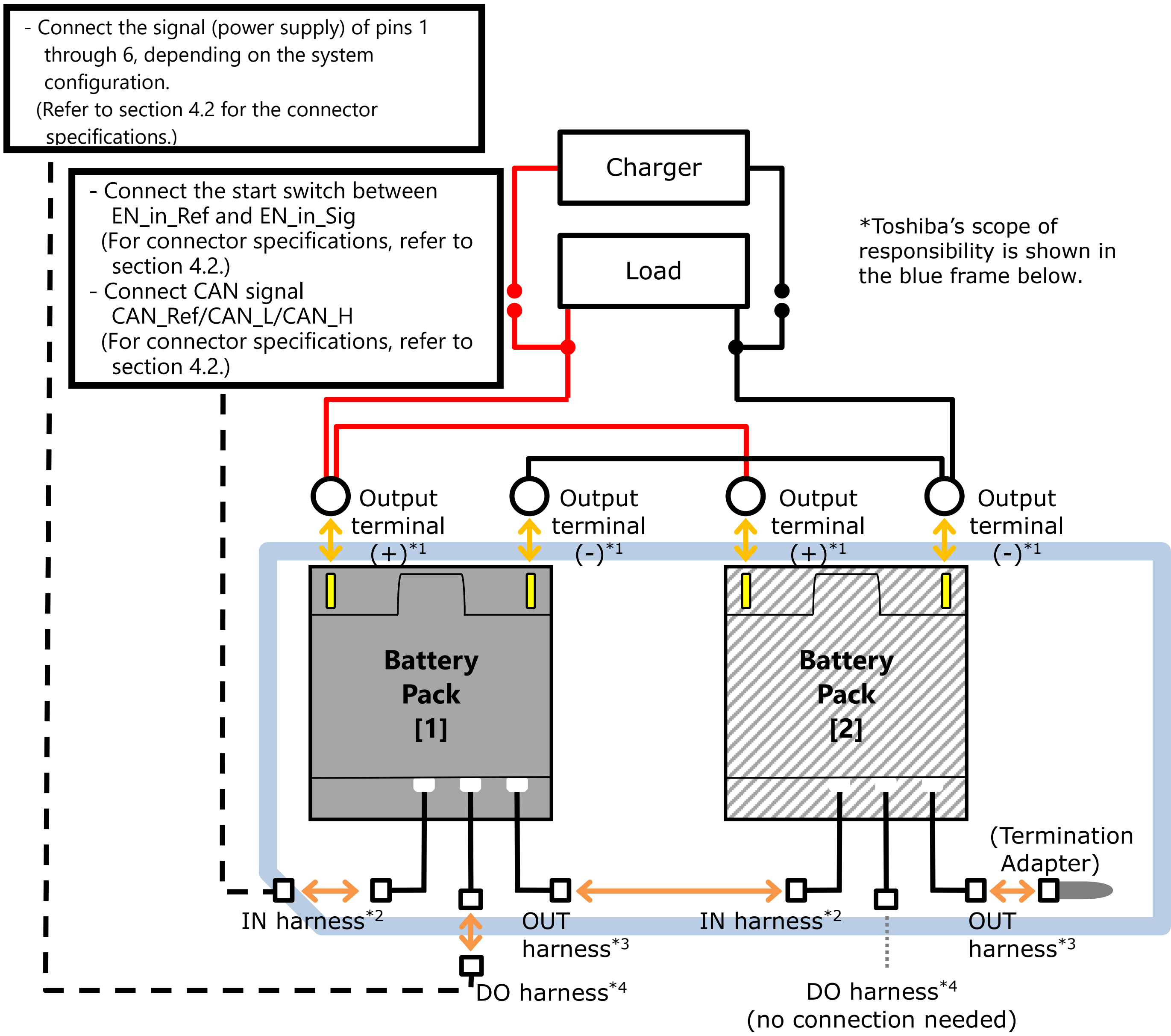


Figure 3-8 Connection Configuration with Both CAN Communication and DO Output

Table 3-8 How to Connect to the Terminal/Connector

	Item	How to connect	Note
*1	Output terminal (+) Output terminal (-)	Tighten with M6 bolts and nuts the cable with M6 round terminal.	Please prepare cables with round terminal and M6 bolts and nuts.
*2	IN harness	Connect harness with connector.	<u>Battery Pack [1]</u> -Please prepare the harness to be connected to termination adapter 2. (Refer to section 4.2 for the connector specifications.) <u>Battery Pack [2]</u> -Connect to OUT harness of Battery Pack [1].
*3	OUT harness	Connect termination adapter.	-
*4	DO harness	Connect harness with connector. *Battery Pack [1] only	Please prepare the harness. Refer to section 4.2 for the connector specifications.

Note: Refer to section 5.4. Protection Function For 2 Packs in Parallel.

4. BMU Specifications

Provided below are the functions of BMU inside the Battery Pack.

4.1. BMU Rated Specifications

Table 4-1 BMU Rated Specifications

Item	Specifications
Rated voltage	25.3 V
Operating voltage range	13.2 V to 33 V
Consumption current	50 mA or less (25 deg C, cell voltage 2.3 V, without cell balance or DO load)
Cell balance current	35 to 45 mA (at cell voltage 2.3 V)
Dark current	200 μA or less (current in the cell, not the connection terminal current)
Protection against moisture	Coating applied to parts where the cell voltage is to be applied

4.2. Signal Harness

Tables 4-2, 4-3, and 4-4 provide information on the applicable connectors and the pin assignment of IN, OUT, and DO harness connectors.

4.2.1. IN Harness

Connector type：JST connector 08R-JWPF-VSLE-D
* Please prepare “JST connector 08T-JWPF-VSLE-D” on the user side.
Cable specification on Battery Pack side： Finished outer diameter 7.0 mm
Minimum bending radius 42 mm (reference value)

Table 4-2 Connector Pin Assignment of IN Harness

Terminal No.	Terminal Name	Note
1	MOD1_in	Pack quantity recognition signal (in) (only to be used for Pack [2] of the two packs in parallel)
2	MOD2_in	
3	ADR1_in	CAN address numbering signal (in) (only to be used for Pack [2] of the two packs in parallel)
4	EN_in_Ref	Activation signal (battery output ON). Activate by short circuiting EN_in_Sig/ Stop by open circuiting*1
5	CAN_Ref	CAN communication and control circuit GND*2
6	CAN_L	CAN communication signal*3
7	CAN_H	
8	EN_in_Sig	Activation signal (battery output ON). Activate by short circuiting EN_in_Ref/ Stop by open circuiting*1

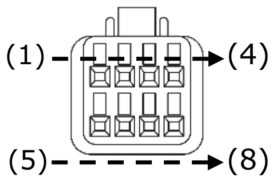
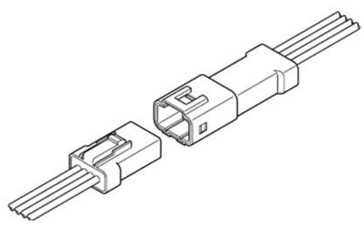


Figure 4-1 08R-JWPF-VSLE-D
Pin Assignment
(Image of Connector Insertion Point)



08T-JWPF-VSLE-D (to the device on the user side)

08R-JWPF-VSLE-D (from the Battery Pack)

Figure 4-2 Image of IN harness connector

*1: Install a contact switch between EN_in_Ref and EN_in_Sig.

If the Battery Pack is not to be used for a long time, turn the contact switch OFF (open circuiting). Leaving the switch ON for a long time may likely lead to overdischarge caused by the power consumption of BMU in the Battery Pack, and render the Battery Pack inoperative.

*2: Caution is needed when grounding the signal line.

There is no conduction between CAN_Ref (terminal No.5; CAN communication and control circuit GND) and the output terminal (-).

*3: When connecting two Battery Packs in parallel without using CAN communication

(i) If the user prepares the terminator

If the user prepares the harness, it is necessary to install CAN terminator. Refer to Figure 4-3, and connect the resistor connect the resistor (120 Ohm ±5%, 1/4 W) between the terminals No.6 (CAN_L) and No.7 (CAN_H).

*The terminator is not necessary when CAN communication is used.

Terminal No.	Signal Name
1	MOD1_in
2	MOD2_in
3	ADR1_in
4	EN_in_Ref
5	CAN_Ref
6	CAN_L
7	CAN_H
8	EN_in_Sig

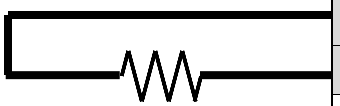


Figure 4-3 Connection of Terminator prepared by User

(ii) If Termination Adapter 2 (model type: FMW-GAA0064P) is used

It is necessary to connect the Termination Adapter 2 to the Battery Pack [1]. Refer to Figure 4-4, and connect the Termination Adapter 2 (refer to section 2.5.3) to the IN harness of the Battery Pack [1]. (See also the connection diagrams in chapter 3. "Connection Configuration.")

The harness to connect the Termination Adapter 2 with the user's device needs to be prepared by the user.

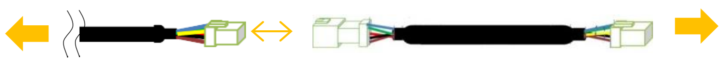


Figure 4-4 Connection of Termination Adapter 2

4.2.2. OUT harness

Connector Type：JST connector 08T-JWPF-VSLE-D
*Connect the Termination Adapter.
Cable specification on Battery Pack side： Finished outer diameter 7.0 mm
Minimum bending radius 42 mm (reference value)

Table 4-3 OUT Harness Connector Pin Assignment

Terminal No.	Terminal Name	Remarks
1	MOD1_out	Pack quantity recognition signal (out)
2	MOD2_out	
3	ADR1_0ut	CAN address numbering signal (out)
4	EN_out_Ref	Activate pack of upper voltage side／Stop signal
5	CAN_Ref	CAN communication and control circuit GND (no conduction with the output terminal (-) of the main circuit)
6	CAN_L	CAN communication signal
7	CAN_H	
8	EN_out_Sig	Activate pack of upper voltage side／Stop signal

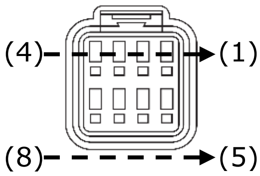


Figure 4-3 08T-JWPFF-VSLE-D
Terminal Position
(Image of Connector Insertion Point)

*Depending on the system configuration, either the Termination Adapter or the IN harness of the Battery Pack [2] is to be connected to the OUT harness.

4.2.3. DO Harness

Connector type: JST 06R-JWPF-VSLE-D
*Please prepare “JST 06T-JWPF-VSLE-D” on the user side.
Cable specification on Battery Pack side: Finished outer diameter 7.0 mm
Minimum bending radius 42 mm (reference value)

Table 4-4 Connector Pin Assignment of DO Harness

Terminal No.	Terminal Item	Note
1	Under voltage warning	Warning to stop discharging
2	Over voltage warning	Warning to stop charging
3	Over temperature warning	Over temperature warning of the battery (cell) or circuit
4 (D04)	SOC1	Indicates SOC condition by 2 Bit.*
5 (D05)	SOC2	
6	5V	25 mA (max.) LED Lighting, etc. Under overload condition above 25 mA, startup failure or CAN communication abnormality may occur. Utmost caution is needed so that the load current should not exceed 25 mA.

*The SOC is outputted as a reference value. The SOC estimation may be inaccurate if the battery has been operating continuously and/or at low temperature.

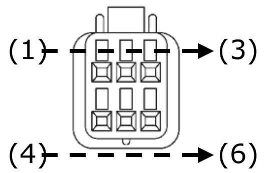


Figure 4-6 06R-JWPF-VSLE-D
Pin Assignment
(Image of Connector Insertion Point)

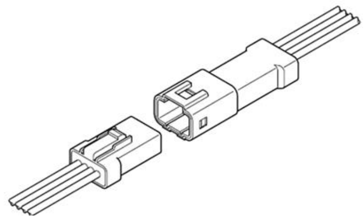


Figure 4-7 DO Harness Connector Image

* The GND for the output signal of DO connector is CAN_Ref (terminal No.5; CAN communication and control circuit GND) of the IN harness. Caution is needed when grounding the signal line.

4.3. Function Specifications

BMU has the following functions:

(1) Cell voltage/current/pack temperature measurement function

(2) Cell voltage balancing function

(3) CAN communication function

(4) DO Output function

(5) Protection function

4.3.1. Cell Voltage/Current/Pack Temperature Measurement Function

BMS built in this Battery Pack measures every cell voltage, current and temperature inside the pack.

4.3.2. Cell Voltage Balancing Function

When a current of 0 A to 10 A is flowing at a cell voltage of 2.1 V or more, the cells are discharged so that the difference between the highest and lowest cell voltages should be 5 mV or less.

4.3.3. CAN Communication Function

The table below shows the specifications of the CAN communication interface with BMU. Control command and the data on the battery pack temperature and voltage, etc. are transmitted from BMU via CAN communication.

Table 4-5 CAN Communication Specification

Item	Details
CAN Protocol Specification	CAN2.0B (Standard format applied) ISO11898-2
Data Format	Standard format (CAN ID:11Bit)
Communication Speed	250kbps
Sample Point	70%

CAN communication output starts approx. 2.4 seconds after the Battery Pack startup (EN switch ON).

4.3.4. DO Output Function

DO output starts approx. 2.4 seconds after the Battery Pack startup (EN switch ON).

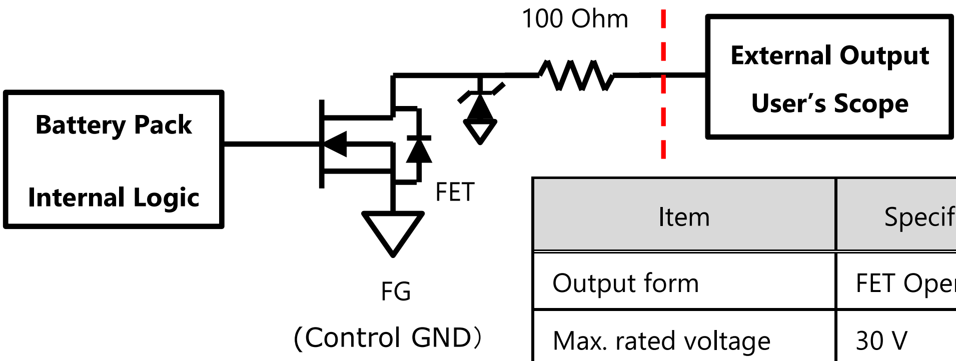
Table 4-6 DO Output Item

<table>
<tr><th>Terminal No.</th><th>Terminal Name</th><th colspan="5">Note</th></tr>
<tr><td>1</td><td>Under voltage warning</td><td colspan="5">Normal: Open(Open-Drain), Abnormal: LOW</td></tr>
<tr><td>2</td><td>Over voltage warning</td><td colspan="5">Normal: Open(Open-Drain) , Abnormal: LOW</td></tr>
<tr><td>3</td><td>Over temperature warning</td><td colspan="5">Over temperature warning of the battery (cell) or circuit
Normal: Open (Open-Drain), Abnormal: LOW</td></tr>
<tr><td rowspan="2">4
(D04)</td><td rowspan="2">SOC1*</td><td colspan="5">Indicates SOC condition by 2 Bit (High=Open-Drain)
Note that the threshold for the output change depends on whether SOC increases or decreases.</td></tr>
<tr><td>Level</td><td>During Charging</td><td>During Discharging</td><td>DO4</td><td>DO5</td></tr>
<tr><td rowspan="4">5
(D05)</td><td rowspan="4">SOC2*</td><td>4</td><td>76% to 100%</td><td>74% to 100%</td><td>Low</td><td>Low</td></tr>
<tr><td>3</td><td>From 51% to below 76%</td><td>From 49% to below 74%</td><td>Low</td><td>High</td></tr>
<tr><td>2</td><td>From 26% to below 51%</td><td>From 24% to below 49%</td><td>High</td><td>Low</td></tr>
<tr><td>1</td><td>From 0% to below 26%</td><td>From 0% to below 24%</td><td>High</td><td>High</td></tr>
<tr><td>6</td><td>5V</td><td colspan="5">25 mA (max.) LED Lighting, etc.
Under overload condition above 25 mA, startup failure or CAN communication abnormality may occur. Utmost caution is needed so that the load current should not exceed 25 mA.</td></tr>
</table>

*The SOC is outputted as a reference value. The SOC estimation may be inaccurate if the battery has been operating continuously and/or at low temperature.

DO Circuit Block

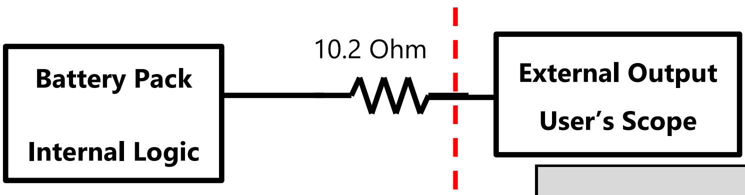
1) Terminal No.1 - No.5 Circuit Block



Item	Specification
Output form	FET Open Drain
Max. rated voltage	30 V
Max. load current	20 mA

Figure 4-8 DO Terminal No.1 - No.5 Circuit Block

2) Terminal No.6 Circuit Block



Item	Specification
Output form	Insulated power supply
Output voltage	4.8 to 5.5 V
Max. load current	25 mA

Figure 4-9 DO Terminal No.6 Circuit Block

In case the 5 V DC is to light up an LED, select one that does not exceed 25 mA.

4.3.5. Protection function

For BMU protection function, refer to section 5.

5. Protection Function

5.1. Overview

Abnormality of the Battery Pack is divided into four stages—warning, error, permanent error, and failure—against which the protection function operates.

The four stages are defined as shown in Table 5-1 below.

Table 5-1 Definitions of Warning/Error/Permanent Error/Failure

Types of Abnormalities	**Definitions of conditions**	**Record to registers**	
		R1	**R2**
Warning	The state leading up to an error	To be recorded	To be recorded
Error	An abnormal state related to cell voltage / temperature, current, or communication	To be recorded	To be recorded
Permanent Error	A seriously abnormal state related to cell voltage / temperature	To be recorded	-
Failure	Malfunction of major components	To be recorded	To be recorded

- R2 register can be cleared by CAN command from the host system.
- R2 register cannot be cleared by Battery Pack power down.
- Permanent Error cannot be cleared.

The above abnormality information of this Battery Pack will be transmitted from the Battery Pack to the host system periodically.

5.2. Timing of DO Output and CAN Notification When Warning or Error is Detected

The timing of DO output and CAN notification of overvoltage, undervoltage, and overtemperature is shown in Figure 5-1 below.

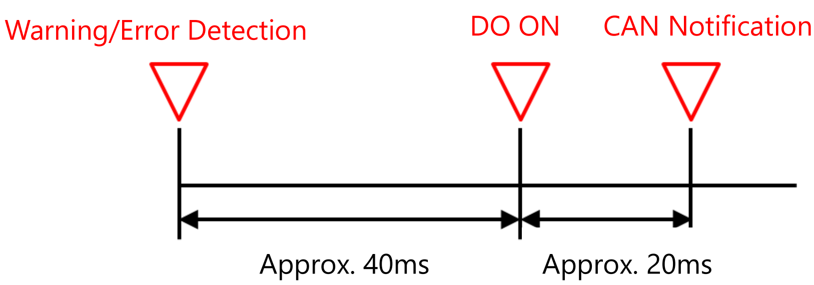


Figure 5-1 Timing of DO Output and CAN Notification When Warning or Error is Detected

5.3. Protection Function of the Battery Pack

For the safe use of the pack, BMU in the Battery Pack is equipped with protection function. The protection function works in accordance with the condition of the Battery Pack.

If the Battery Pack comes to a stop, wait a while and try re-activating it.

If the Battery Pack fails to be re-activated, it may be in the state of “Permanent Error” or “Failure” as defined in the table below.

*For details, refer to section 7.7 of the instruction manual (SPC-COM-E0063).

Table 5-2 Battery Pack Protection Item List (Warning)

Item	Judgment Condition	Pack 0peration After Judgment							Expected Host System Operation After Judgment	Cancellation Condition
		Charge FET	Discharge FET	Shut down	CAN Communication		DO Output			
					bit	Output	Terminal	Output		
Over voltage warning	Max. cell voltage in the pack is 2.7 V or higher[*1]	ON	ON	–	b7	Output	No.2	Low	Stop charging. Discharge.	Max. cell voltage in the pack is 2.55 V or lower.
Under voltage warning	Min. cell voltage in the pack is 1.5 V or lower[*1]	ON	ON	–	b6	Output	No.1	Low	Stop discharging. Charge.	Min. cell voltage in the pack is 2.1 V or higher
	Min. cell voltage in the pack is 1.9 V or lower[*2]								Cancel the state that the pack is left with startup (SW-off and shutdown)	
Pack over temperature warning	Pack temperature is 55 degC or higher[*1]	ON	ON	–	b3	Output	No.3	Low	Stop charging/ discharging.	Pack temperature is 50degC or lower
Pack under temperature warning	Pack temperature is -30 degC or lower[*1]	ON	ON	–	b2	Output	–	–	Stop charging/ discharging.	Pack temperature is -25degC or higher
Circuit over temperature warning	FET or shunt temperature (higher one) is 90 degC or higher	ON	ON	–	b1	Output	No.3	Low	Stop charging/ discharging.	FET or shunt temperature (higher one) is 85 degC or lower
Cell voltage difference warning	(Max. cell voltage) – (Min. cell voltage) in the pack is 500 mV or higher.	ON	ON	–	b0	Output	–	–	Stop charging/ discharging.	(Max. cell voltage) – (Min. cell voltage) in the pack is 400mV or lower.

*1: The judgment time is 400 to 600 msec. And it takes another 200 msec (typ.) to notify the host system via CAN.
*2: The judgment time is 300 sec. And it takes another 200 msec (typ.) to notify the host system via CAN.

Table 5-3 Battery Pack Protection Item List (Error)

Item	Judgment Condition	Pack Operation After Judgment: Charge FET	Discharge FET	Shut down	CAN Communication: bit	CAN Communication: Output	DO Output: Terminal	DO Output: Output	Expected Host System Operation After Judgment	Cancellation Condition
Over voltage error	Max. cell voltage in the pack is 2.8 V or higher[*1]	OFF	ON	–	b7	Output	No.2	Low	Discharge	Max. cell voltage in the pack is 2.55 V or lower.
Under voltage error	Min. cell voltage in the pack is 1.4 V or lower[*1]	ON	OFF[*5]	Shut down[*2]	b6	Output	No.1	Low	Charge	Min. cell voltage in the pack is 2.1 V or higher.
	Min. cell voltage in the pack is 1.9 V or lower[*6]		OFF							
Charge over current error	Charge current remains ・125 A or higher for 220 sec ・210 A or higher for 2 sec ・30 A or higher[*3]	OFF	OFF	Shut down	b5	Output	–	–	–	The pack is shut down.
Discharge over current error	Discharge current remains ・125 A or higher for 220 sec ・210 A or higher for 2 sec	OFF	OFF	Shut down	b4	Output	–	–	–	The pack is shut down.
Pack over temperature error	Pack temperature is 66degC or higher[*1]	OFF	OFF	Shut down	b3	Output	–	–	–	The pack is shut down.
Pack under temperature error	Pack temperature is -35degC or lower[*1]	OFF	OFF	Shut down	b2	Output	–	–	–	The pack is shut down.
Circuit over temperature error	FET or shunt temperature (higher one) is 95degC or higher[*1]	OFF	OFF	Shut down	b1	Output	–	–	–	The pack is shut down.
Pack voltage difference error	Voltage difference between two packs in parallel remains above 0.3 V for 1 sec at startup	OFF[*5]	OFF	–	b0	Output	–	–	–	The voltage difference is 0.3 V or lower, and the current is less than 30 A.

*1: The judgment time is 400 to 600 msec. And it takes another 200 msec (typ.) to notify the host system via CAN.
*2: The pack is shut down if the pack cannot confirm charge current above 1.6 A within 60 sec.
*3: The charge current of 30 A or higher is judged to be a charge over current error in case the two packs in parallel are having a pack voltage difference error.
*4: Of the two packs in parallel, only the one with the lower voltage is to have its charge FET turned on and is able to be charged.
*5: If under voltage error is detected just after starting, discharge FET is ON. If the state which min. cell voltage has decreasing 20mV lasts for 10 sec or more since the error had confirmed, discharge FET transfers to OFF.
*6: The judgment time is 600 sec. And it takes another 200 msec (typ.) to notify the host system via CAN.

Table 5-4 Battery Pack Protection Item List (Permanent Error)

Item	Judgment Condition	Pack Operation After Judgment							Expected Host System Operation After Judgment	Cancellation Condition
		Charge FET	Discharge FET	Shut down	CAN Communication		DO Output			
					bit	Output	Terminal	Output		
Over voltage permanent error	Max. cell voltage in the pack is 3.0 V or higher*1	OFF	OFF	—	b7	Output	No.2	Low	Stop use.	Unable to cancel
Under voltage permanent error	Min. cell voltage in the pack is 1.2 V or lower*2	OFF	OFF	—*3	b6	—*4	—	—	Stop use.	Unable to cancel

*1: The judgment time is 400 to 600 msec. And it takes another 200 msec (typ.) to notify the host system via CAN bus.
*2: The judgment time is 1800 to 2000 msec. And it takes another 200 msec (typ.) to notify the host system via CAN bus.
*3: The Battery Pack enters into a state of shutdown in Error (refer to Table 5-3), so no instruction for shutdown is given for permanent error.
*4: Although the control inside the battery is set up so that CAN is able to be outputted, CAN is in fact not outputted because the battery is shut down in the state of permanent error.

Table 5-5 Battery Pack Protection Item List (Failure 1)

Item	Judgment Condition	Pack Operation After Judgment					Cancellation Condition
		Charge FET	Discharge FET	Shut down	CAN Communication		
					bit	Output	
MPU failure (self diagnosis)	Error related to MPU is detected by the self-diagnosis of pack	OFF	OFF	—	b7	*1	The pack is shut down.
VTM communication failure	Error related to VTM communication is detected by the self-diagnosis of pack	OFF*2	ON	—	b6	Output	The pack is shut down.
Pack misconnection failure	Error is detected by the pack connection which does not match with the number of packs.	OFF	OFF	—	b5	Output	The pack is shut down.
Pack configuration Failure	Error is detected by the pack connection which does not match with the connection type. (Unconnected terminators, etc)	OFF	OFF	—	b4	Output	The pack is shut down.

*1: It depends on where the failure takes place.
*2: Charge FET is forced OFF 1 minute after the failure notification is sent via CAN. However, if any of the conditions below is satisfied, Charge FET is forced OFF immediately.
- Charge current of 15 A or more continues to flow for 2 sec.
- The integrated value of the charge current has increased by 0.1 Ah or more after the failure occurrence.
- Current measurement circuit failure and cell voltage measurement failure are detected at the same time.

Table 5-6 Battery Pack Protection Item List (Failure 2)

Item		Judgment Condition	Pack Operation After Judgment					Cancellation Condition
			Charge FET	Discharge FET	Shut down	CAN Communication		
						bit	Output	
VTM failure (temp. monitor)		Error related to VTM temperature measurement function is detected by the self-diagnosis of pack	OFF*1	ON	–	b7	Output	The pack is shut down.
Current measurement circuit failure		Error related to current sensing circuit is detected by the self-diagnosis of pack	OFF*1	ON	–	b6	Output	The pack is shut down.
EEPROM failure (important area)		Error related to important information stored in EEPROM is detected by the self-diagnosis of pack	OFF*1	ON	–	b5	Output	The pack is shut down.
CAN Communication failure between packs	1 pack alone	N/A	ON	ON	–	b4	–	N/A
	Between 2 packs in parallel	Error is detected when the pack fails to receive CAN communication or normal data from the other pack for 2 seconds.	OFF	ON	Shut down*3	b4	Output	The pack is shut down.
VTM failure (voltage monitor)		Error related to VTM voltage measurement function is detected by the self-diagnosis of pack	OFF*1	ON	–	b2	Output	The pack is shut down.
Failure of circuit temperature measurement		Error related to circuit temperature measurement function is detected by the self-diagnosis of pack	OFF*1	ON	–	b1	Output	The pack is shut down.
AD conv. failure		Error related to AD conv. is detected by the self-diagnosis of pack	OFF*1	ON	–	b0	Output	The pack is shut down.

*1: Charge FET is forced OFF 1 minute after the failure notification is sent via CAN. However, if any of the conditions below is satisfied, Charge FET is forced OFF immediately.

- Charge current of 15 A or more continues to flow for 2 sec.
- The integrated value of the charge current has increased by 0.1 Ah or more after the failure occurrence.
- Current measurement circuit failure and cell voltage measurement failure are detected at the same time.

*2: The Battery Pack is shut down if the failure continues for 30 min. or more.

Table 5-7 Battery Pack Protection Item List (Failure 3)

Item	Judgment Condition	Pack Operation After Judgment					Cancellation Condition
		Charge FET	Discharge FET	Shut down	CAN Communication		
					bit	Output	
VTM failure (except for voltage and temp. monitor)	Error related to VTM function except for voltage and temp. monitor is detected by the self-diagnosis of pack	ON	ON	–	b6	Output	The pack is shut down.
EEPROM failure (normal area)	Error unrelated to important data stored in EEPROM is detected by the self-diagnosis of pack	ON	ON	–	b4	Output	The pack is shut down.

5.4. Protection Function For 2 Packs in Parallel

The Battery Pack is equipped with protection function which prevents the flow of excessive electric current in case the voltage difference exceeds 0.3 V between 2 packs connected in parallel (Discharge FET: OFF, Charge FET: ON). Note that, **if the voltage difference is 0.3 V or more, the following restrictions are applied until the voltages are equalized.**

- Voltage and signals cannot be outputted.
- The Battery Pack can only be charged at a charge current of 25 A or less.
- Of the 2 packs, only the one with the lower voltage can be charged.

If the restrictions are applied, be sure to fully charge the Battery Packs in CC/CV-charging conditions shown below.

CC-current: **1 A or more, 25 A or less**
CV-voltage: **28.6 V**

Figure 5-2 below is the protection flow for 2 Battery Packs connected in parallel. It is necessary to have the voltages equalized as instructed with reference to the precautions above.

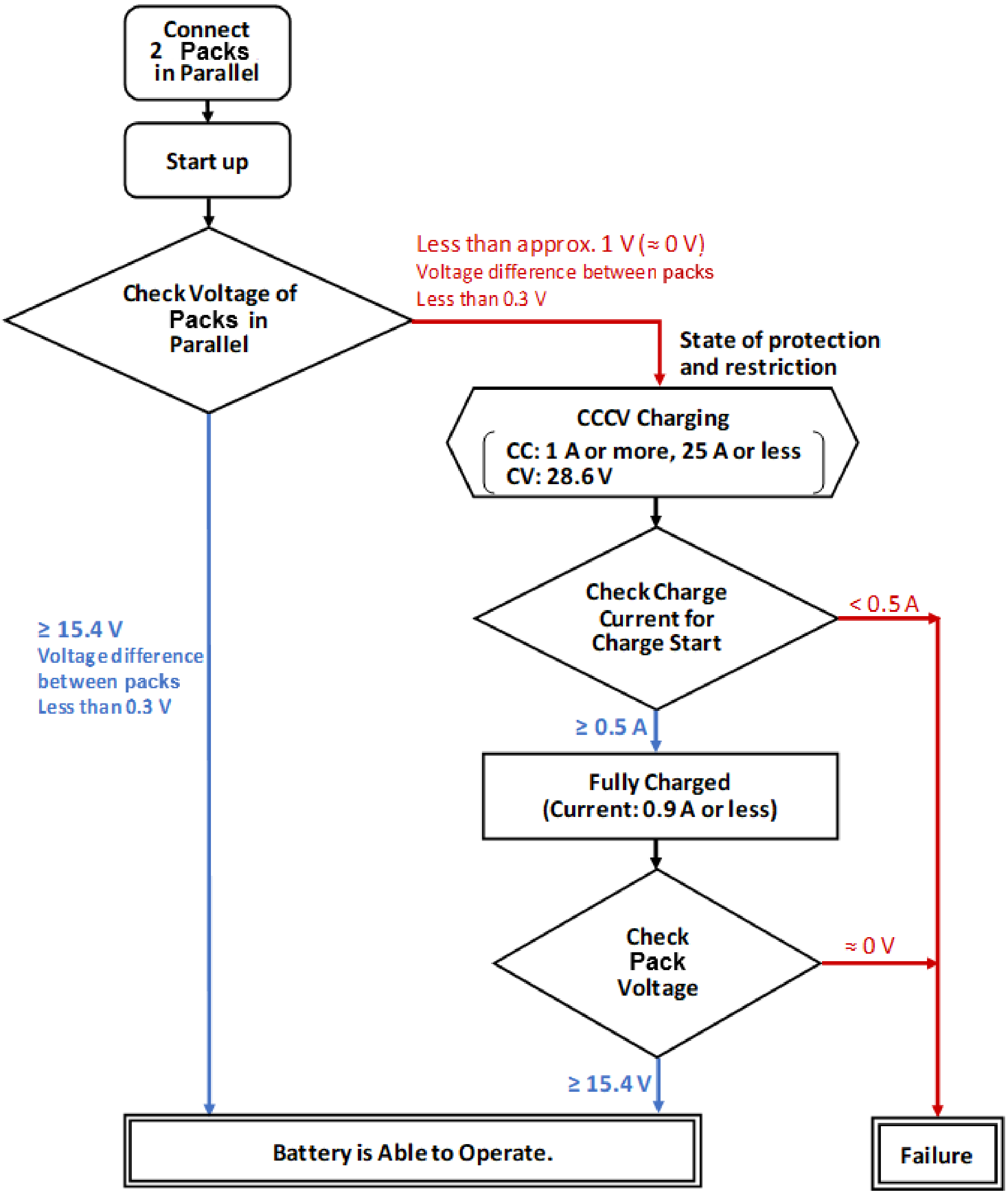


Figure 5-2 Protection Flow for 2 Battery Packs Connected in Parallel

6. Cautions on Use

6.1. Precautions on Charging

(1) Before turning on the system, check the connection status of the connected devices.

(2) When charging the Battery Pack, be sure to follow the instructions below.

- A warning will notify you of over voltage through DO and CAN communication when either of the cell voltage reaches 2.7 V. Stop charging when you get the warning.
- Make sure that the charge voltage is 29.7 V or less.
- Although the Battery Pack startup in the state of low SOC (approximately 15.4 V or less; min. cell voltage 1.2 to 1.4V), it outputs power so that the charger can check the battery voltage. The purpose of the function is monitoring the voltage. If charging is not checked until the cell voltage decreases 20 mV or more by discharging or current consumption, or 1 min. has elapsed since startup, the Battery Pack stops discharging and be shut down. (It is applicable to packs of which Serial No. is started with "G123".)

<u>The recommended charging conditions are shown below.</u>

- Charging control:	CC/CV control (Constant Current/ Constant Voltage control) Do not use a charger which only has a voltage control system. Use of a charger with no current control system will result in over current charging, and fail to charge the Battery Pack appropriately.
- Charge voltage:	28.6 V or less (for constant-current charging, voltage cut charging) 28.6 V (for constant voltage for constant-current constant-voltage charging)
- Protection on the charger:	To prevent overcharge or accidents caused by charging a broken Battery Pack, select a charger which allows the maximum and minimum voltage limits to be set as shown below, and which stops charging when the limits are exceeded. Maximum voltage limit: 29.7 V or less. Minimum voltage limit: 13.2 V or more. Note: If the battery voltage fails to rise after charging, there may be abnormalities with the battery or the charging path. If this happens, you are advised to stop the charging.
- Charge current:	125 A (maximum) (within the range where the cell temperature does not exceed 55deg C)
- Emergency stop of charging:	You are advised to select a charger that, if abnormality is detected on the Battery Pack during charging, can receive the abnormality information and stop its output.
- Operating ambient temperature:	-30 to 45 deg C
- Humidity:	85% RH or lower (with no condensation)

*If you consider charging conditions other than those described above, please contact us at the contact information provided in section 8.6. of the instruction manual (SPC-COM-E0063).

(3) Besides the above-mentioned precautions, follow those described in section 7.4 of the instruction manual (SPC-COM-E0063).

6.2. Precautions on Discharging

(1) Before turning on the system, check the connection status of the connected devices.

(2) When discharging the Battery Pack, please follow the instructions below.

- A warning will notify you of under voltage through DO and CAN communication when either of the cell voltage drops to 1.5 V. Stop discharging when you get the warning.

-Make sure that the discharge voltage of the Battery Pack does not drop to 16.5 V or less.

- If the Battery Pack is left uncharged after reaching the discharge cut-off voltage (16.5 V), it will reach the under voltage error. Be sure to charge the pack within one hour after its reaching the discharge cut-off voltage.

- If the Battery Pack is left uncharged after reaching under voltage error, it will fall into the state of under voltage permanent error and never be able to be used again. Be sure to charge the pack within one week after its reaching under voltage error.

(3) Besides the above-mentioned precautions, follow those described in section 7.4 of the instruction manual (SPC-COM-E0063).

7. Contact

For the inquiry about the product, please contact to the distributer of the product.

<Contact of manufacturer>
Toshiba Corporation, Battery Div.
72-34, Horikawa-cho, Saiwai-ku Kawasaki-shi, Kanagawa 212-8585, Japan
https://www.webcom.toshiba.co.jp/scib/en/contact.php

For frequently asked technical questions, please see the link below.
https://www.scib.jp/en/product/sip/faq-tech.php

Toshiba Corporation

February 22th, 2017 : New issue

October 15th, 2025 : Eighth edition